



ACKNOWLEDGMENTS

The publisher would like to thank the following people for their help with making the book: Bharti Bedi, Upamanyu Das, Arpita Dasgupta, Priyanka Kharbanda, Sai Prasanna, Bipasha Roy, Anuroop Sanwalia, Shambhavi Thatte, and Vatsal Verma for editorial assistance; Rakesh Kumar, Tanya Mehrotra, Priyanka Sharma, and Saloni Singh for the jacket; and Jo Penning for proofreading and indexing.

Consultants at the Smithsonian Institution:

National Air and Space Museum:
Bob Van der Linden, Curator of Commercial Aircraft, Aeronautics
Matt Shindell, Curator of Planetary Science and Exploration, Space
Michael Neufeld, Curator of Rockets and Missiles, Space

National Museum of Natural History:
Carole Baldwin, Curator of Fishes, Department of Vertebrate Zoology
Benjamin Andrews, Research Geologist, Director
Global Volcanism Program Mineral Sciences

Smithsonian Enterprises:

Kealy Gordon, Product Development Manager
Jill Corcoran, Director of Licensed Publishing
Janet Archer, DMM Ecom and D-T-C
Carol LeBlanc, President, SE

Picture Credits

The publisher would like to thank the following for their kind permission to reproduce their photographs:

(Key: a-above; b-below/bottom; c-center; f-far; l-left; r-right; t-top)

1 Getty Images: Pool (c). **2-3 Science Photo Library:** Tony & Daphne Hallas. **4 Getty Images:** Alexis Rosenfeld (ca). **NASA:** NASA / JPL-Caltech (cra). **5 Alamy Stock Photo:** Science Photo Library (cla); Jim West (ca). **6-7 Alamy Stock Photo:** Daryl Mulvihill. **8 Shutterstock.com:** Kimimasa Mayama / Epa-Efe / Shutterstock (bl). **12-13 Getty Images:** The Asahi Shimbun. **12 Getty Images:** Andreas Solaro (cra). **14-15 Octinion.** **14 Alamy Stock Photo:** ZUMA Press, Inc. (c). **16-17 Alamy Stock Photo:** Conrad Elias. **18-19 Cephas Picture Library:** Mick Rock. **18 Alamy Stock Photo:** agefotostock (tl); Jake Lyell (cl); Photology1971 (bl). **19 Alamy Stock Photo:** Farlap (c). **20-21 Getty Images:** Handout. **21 SuperStock:** Biosphoto (c). **22-23 NASA.** **22 NASA:** (c). **24-25 Ciril Jazbec.** **24 Alamy Stock Photo:** OA (cra). **26-27 Peter Trautwein Aqualonis.** **26 Peter Trautwein Aqualonis:** (crb). **28-29 Shutterstock.com:** Ksenia Ragozina. **28 Alamy Stock Photo:** Phil Rees (c). **30-31 Nat Geo Image Collection:** Greg Ludwig. **30 Alamy Stock Photo:** Cavan Images (c). **32-33 Getty Images / iStock:** Chunyip Wong. **32 Shutterstock.com:** Anongnaj Phewngern (cl). **34-35 Shutterstock.com:** urbans. **36-37 Shutterstock.com:** polu_tsvet. **37 Alamy Stock Photo:** H. Mark Weidman Photography (tc). **38-39 Massachusetts Institute of Technology (MIT):** Bob Mumgaard. **40-41 NASA:** NASA / Roscosmos. **40 Alamy Stock Photo:** dpa picture alliance archive (bc). **41 Alamy Stock Photo:** All Canada Photos (bc). **Science Photo Library:** Trevor Clifford Photography (bl). **42-43 Neoen:** (c). **42 Alamy Stock Photo:** Askar Karimullin (c). **44-45 Getty Images:** Loic Venance. **44 Getty Images:** Fred Tanneau (tl). **45 Science Photo Library:** Martin Bond (c). **46-47 Vattenfall Group.** **48-49 Getty Images / iStock:** greenleaf123. **48 Alamy Stock Photo:** Aflo Co. Ltd. (clb). **49 Dreamstime.com:** Bencharat Chanphong (c). **52-53 Boeri Studio:** Vertical Forest Ph. Dimitar Harizanov, Milan, Italy. **52 Studio Roosegaarde:** (cl). **54-55 Alamy Stock Photo:** Ivan Smuk. **55 Alamy Stock Photo:** Dennis Frates (cr). **56-57 BAS:** British Antarctic Survey / Antony Dubber.

58-59 Getty Images: Ventdusud. **59 Alamy Stock Photo:** Imaginechina Limited (c). **60-61 Getty Images:** VCG. **60 Shutterstock.com:** guruXOX (c). **61 Getty Images:** VCG (tr). **62-63 Getty Images:** Alexis Rosenfeld. **64-65 Alamy Stock Photo:** David Tadevosian. **64 Alamy Stock Photo:** Science Photo Library (c). **66-67 ESO:** ESO / F. Kamphues. **66 Alamy Stock Photo:** Andrey Armyagov (cl); ZUMA Press, Inc. (tl). **Getty Images:** Arctic-Images (bl). **68 Reuters:** Navesh Chitrakar. **Science Photo Library:** Eye Of Science (clb). **69 Alamy Stock Photo:** EyeSee Microstock. **Dreamstime.com:** Martin Kawalski (c). **70-71 Alamy Stock Photo:** Lev Karavanov. **71 Shutterstock.com:** Valokuva24 (tr). **72-73 Getty Images:** Anadolü Agency. **73 Depositphotos Inc.:** Gudkovandrey (c). **74-75 Alamy Stock Photo:** Dino Fracchia. **74 Dreamstime.com:** Rdonar (tl). **76-77 Alamy Stock Photo:** CW Images. **76 Getty Images:** Jan-Otto (c). **77 Alamy Stock Photo:** Reuters (tr). **Dreamstime.com:** Singhsomendra (br). **Getty Images:** Jonas Gratzer (cr). **78-79 Getty Images:** Bloomberg. **78 Getty Images:** Star Tribune via Getty Images (c). **79 Alamy Stock Photo:** Frances Roberts (tl). **Getty Images:** Monty Rakusen (tr). **SuperStock:** Michael Rosenfeld / Maximilian S (tc). **80-81 Getty Images:** Patrick AVENTURIER. **82-83 Action Plus Sports Images:** Imago / Actionplus. **82 Alamy Stock Photo:** Arterra Picture Library (cl). **Getty Images:** George Rose (tl). **Shutterstock.com:** oNabby (bl). **84-85 Dreamstime.com:** Vitalyedush / All. **88-89 Extreme E.** **89 NASA:** (crb). **90-91 Roborace:** Robocar 1.0 (c). **92 Alamy Stock Photo:** riddypix (tr). **94-95 Alamy Stock Photo:** Thom Lang. **94 123RF.com:** foottoo (c). **95 Getty Images:** MarBom (tr). **96-97 Getty Images:** Pool. **97 Getty Images:** Stringer (cr). **Shutterstock.com:** Danny Ecker (br); Konstantin Tronin (tr). **98-99 Alamy Stock Photo:** dpa picture alliance. **98 Science Photo Library:** Dr. Gary Settles (cl). **100-101 Dorling Kindersley:** Hum3D.com (c). **TurboSquid:** 3d_molier International (Astronaut). **101 Alamy Stock Photo:** RGB Ventures / SuperStock (c). **102-103 Getty Images:** Roscosmos Press Office. **102 Alamy Stock Photo:** ITAR-TASS News Agency (tl). **103 Alamy Stock Photo:** Robert Taylor (c). **104-105 NASA:** NASA / JPL-Caltech. **104 NASA:** NASA / JPL-Caltech (cla). **105 Science Photo Library:** Detlev Van Ravenswaay (br). **106-107 Shutterstock.com:** Ashley Cooper / Specialiststock / Splashdown. **108-109 Getty Images:** Christopher Kimmel / Aurora Photos. **110-111 Alamy Stock Photo:** AB Forces News Collection. **111 Alamy Stock Photo:** Jack Sullivan (tr). **112 Bryan Christie Design:** (cl). **112-113 Getty Images:** The Asahi Shimbun. **113 Getty Images:** VCG (tr). **Photo by courtesy of Seatools:** (cr, br). **114-115 Alamy Stock Photo:** Prisma by Dukas Presseagentur GmbH. **115 Dreamstime.com:** Photoeuphoria (c). **116-117 Science Photo Library:** James King-Holmes. **117 Alamy Stock Photo:** Andriy Popov (br). **Getty Images:** Bloomberg (cr). **Science Photo Library:** Mauro Fermariello (tr). **118-119 Getty Images:** Bartosz Siedlik. **119 Alamy Stock Photo:** Aleksey Boldin (c). **122-123 Science Photo Library:** Wim Van Egmond. **122 Getty Images:** BSIP (clb). **124-125 Science Photo Library:** Biophoto Associates. **126 Alamy Stock Photo:** Trevor Smith (cl). **126-127 Alamy Stock Photo:** Science Photo Library. **128-129 Getty Images:** SDI Productions. **128 Alamy Stock Photo:** Science Photo Library (c). **129 Alamy Stock Photo:** Jiri Hubatka (tc). **130-131 Science Photo Library:** Medical Media Images. **130 Getty Images:** Peter Dazeley (c). **132-133 CYBERDYNE:** Prof. Sankai University of Tsukuba / CYBERDYNE Inc.. **132 Shutterstock.com:** (bl). **134-135 Science Photo Library:** Burger / Phanie. **135 Alamy Stock Photo:** BSIP SA (br); Wavebreak Media Ltd. (tr). **Science Photo Library:** GARO / PHANIE (cr). **137 Getty Images:** Jim Watson (c). **138-139 Getty Images:** Kyodo News. **138 Getty Images:** The Washington Post (crb). **140-141 Getty Images:** Bertrand Guay. **140 Getty Images:** (cra). **142-143 Getty Images:** Boston Globe. **143 Alamy Stock Photo:** Nature Picture Library (tr). **Getty Images / iStock:** Vladimir

Vladimirov (cr). **Science Photo Library:** (crb). **144-145 Getty Images:** The AGE. **145 NASA:** (cr). **146-147 NOAA:** Collection of Doug Helton, NOAA / NOS / ORR. **147 Getty Images:** Martin Harvey (cr). **148-149 Getty Images:** Nigel Hicks / Eden Project. **148 Biosphere 2:** The University of Arizona (c). **149 Getty Images:** Franz-Marc Frei / Eden Project (crb). **150-151 Alamy Stock Photo:** Matthew Oldfield Underwater Photography. **150 Dreamstime.com:** Jan Lorenz (tl). **152-153 Alamy Stock Photo:** David Fleetham. **154-155 Science Photo Library:** Tony & Daphne Hallas. **154 Getty Images / iStock:** Georgeclerk (clb). **156-157 Getty Images:** NurPhoto. **156 Alamy Stock Photo:** Reuters (tl). **157 Alamy Stock Photo:** Blickwinkel (c). **160-161 Nick Verola.** **161 Getty Images:** Lev Fedoseyev (clb). **162-163 Caters News Agency:** Geoff Mackley. **163 Alamy Stock Photo:** UPI (c). **164 Alamy Stock Photo:** XM Collection (c). **166-167 NOAA:** ThayerMahan, Inc. Kraken Robotics, and the NOAA Office of Ocean Exploration and Research. **166 Alamy Stock Photo:** Thomas Imo (crb). **168-169 Nat Geo Image Collection:** Carsten Peter. **169 Alamy Stock Photo:** Jim West (cr). **Science Photo Library:** British Antarctic Survey (tr, crb). **170-171 Yuya Makino, IceCube/National Science Foundation.** **171 Alamy Stock Photo:** Image Source (cra). **172-173 Gran Telescopio Canarias:** Pablo Bonet / Instituto de Astrofísica de Canarias. **173 Alamy Stock Photo:** Imaginechina Limited (bc). **174-175 NASA:** NASA, ESA / Hubble and the Hubble Heritage Team. **176-177 Science Photo Library:** Ikelos GmbH / Dr. Christopher B. Jackson. **176 123RF.com:** Sergeychayko (tl). **Science Photo Library:** Clouds Hill Imaging Ltd. (bl); IKELOS GMBH / DR. CHRISTOPHER B. JACKSON (cl). **177 Getty Images:** Valery Sharifulin (c). **178-179 Getty Images:** Patrick Aventurier. **180-181 Alamy Stock Photo:** PA Images. **180 Science Photo Library:** Alexander Tsias (cl). **182-183 Science Photo Library:** Pascal Goetgheluck. **182 Science Photo Library:** Philippe Plailly (cl). **183 Kennis & Kennis / Alfons and Adrie Kennis:** (br, bc, bl). **184-185 Alamy Stock Photo:** Maximilian Buzun. **184 Dreamstime.com:** Petr Taborsky (c). **185 Alamy Stock Photo:** Ivan Smuk (tc). **Getty Images:** Contributor (crb); Justin Sullivan (cra). **Getty Images / iStock:** undefined undefined (cr). **186-187 John Downer Productions.** **186 Getty Images:** Yamaguchi Haruyoshi (crb). **187 John Downer Productions:** (br). **188-189 Shutterstock.com:** Kimimasa Mayama / Epa-Efe / Shutterstock. **190-191 Getty Images:** Monty Rakusen. **190 Alamy Stock Photo:** Martin Shields (c). **191 Alamy Stock Photo:** Roibu (cr); Science Photo Library (tr, br). **192-193 Science Photo Library:** Du Cane Medical Imaging Ltd. **192 Massachusetts Institute of Technology (MIT):** Rebecca Saxe, Ben Deen, Atsushi Takahashi / Department of Brain and Cognitive Sciences, M (c). **194-195 Science Photo Library:** Klaus Guldbrandsen. **194 Science Photo Library:** Chris Priest (c). **196-197 Alamy Stock Photo:** Mike Lane. **196 naturepl.com:** Ingo Arndt (crb). **198-199 Nat Geo Image Collection:** Steve Winter. **199 Getty Images:** The Washington Post (c). **200 Science Photo Library:** Medical Media Images (tr). **Shutterstock.com:** polu_tsvet (tc). **201 Alamy Stock Photo:** Dino Fracchia (tr). **Getty Images / iStock:** Georgeclerk (tc); greenleaf123 (ftl). **NASA:** (tl). **202 Alamy Stock Photo:** Ivan Smuk (tl). **Getty Images:** VCG (tr). **Science Photo Library:** James King-Holmes (tc). **Shutterstock.com:** Konstantin Tronin (ftl). **203 Alamy Stock Photo:** dpa picture alliance archive (tl). **ESO:** ESO / F. Kamphues (tr). **Getty Images:** Bloomberg (ftl); MarBom (tc). **204 Alamy Stock Photo:** Maximilian Buzun (tr); David Tadevosian (tc). **205 Getty Images:** BSIP (tl). **Getty Images / iStock:** Vladimir Vladimirov (tr). **Science Photo Library:** Burger / Phanie (ftl); Philippe Plailly (tc).

All other images © Dorling Kindersley

For further information see:
www.dkimages.com